

SAI SRI RAM MARNENI

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PROFESSIONAL SUMMARY

AI Engineer with 2 years of industry experience building LLM-powered systems, multi-agent orchestration frameworks, and RAG pipelines at Mphasis. Currently completing an MSc in Artificial Intelligence at Munster Technological University (graduating Aug 2026). Hands-on with Python, CrewAI, LangChain, FastAPI, and Model Context Protocol (MCP) integrations. Experienced in taking agentic systems from prototype to deployment, with a focus on runtime reliability, adaptive feedback loops, and privacy-preserving AI. Targeting Applied AI Engineer and Graduate AI Engineer roles in Ireland.

SKILLS

Languages: Python, SQL, JavaScript

AI / ML: LLMs, Retrieval-Augmented Generation (RAG), NLP, Generative AI, Deep Learning, Scikit-learn, Prompt Engineering

Frameworks: CrewAI, LangChain, FastAPI, Streamlit, HTTPX

Cloud / Tools: Azure OpenAI, OpenAI API, ChromaDB, MongoDB, PostgreSQL, Git

Concepts: Multi-Agent Systems, Model Context Protocol (MCP), Vector Databases, API Orchestration, Runtime Feedback, Agentic AI

EXPERIENCE

Associate Software Engineer | Mphasis, Bengaluru

Aug 2023 – Aug 2025

- Built and deployed a 5-agent supply chain automation system using CrewAI and Model Context Protocol (MCP), enabling AI-driven decision workflows across multi-turn enterprise interactions with SQL-backed data retrieval.
- Improved employee turnover classification model accuracy from ~45% to ~78% by applying EDA, SMOTE class balancing, k-fold cross-validation, clustering analysis, and systematic hyperparameter tuning — identifying the best-performing model and surfacing actionable retention strategies.
- Developed a privacy-preserving content moderation system using locally hosted LLMs (LLaMA 2, BERT, Mistral, GPT-3.5) to assess input text for harm and relevance in real time, eliminating reliance on third-party APIs and laying the foundation for future plugin integrations.
- Engineered Python-based LLM solutions and internal AI tooling that automated data pipeline workflows, reducing manual processing effort and improving pipeline reliability across the team.

PROJECTS

Self-Adaptive Memory and Feedback Orchestrator (SAMFO)

2025 – 2026 | Thesis

Python, CrewAI, FastAPI, HTTPX, JSON

- Designed and built a self-adaptive multi-agent orchestration framework coordinating 4–5 agents with shared memory, runtime observability, and adaptive API reliability — submitted as MSc thesis at MTU.
- Implemented real-time schema drift detection, timeout handling, and rule-based API failover logic, significantly reducing failure propagation across agent-to-API communication layers.
- Validated the framework under controlled failure scenarios and live API conditions, demonstrating improved resilience and observability compared to static orchestration approaches.

Multi-Agent RAG and Contract Lifecycle Management (CLM)

2024

Python, LangChain, OpenAI, CrewAI, ChromaDB, Streamlit

- Built a multi-agent Retrieval-Augmented Generation (RAG) pipeline enabling intelligent semantic search over a corpus of 50–200 PDF contracts and documents using LangChain, OpenAI embeddings, and ChromaDB vector storage.
- Optimised vector database retrieval performance through embedding evaluation and index tuning, improving query relevance across contract data.

Multi-Agent Supply Chain Management System

2024

Python, CrewAI, PostgreSQL, MCP

- Developed a 5-agent supply chain system supporting multi-turn interactions and SQL-driven backend data retrieval, with MCP integration and synthetic dataset creation for AI-powered operational decision workflows.
- Employee Turnover Analysis** 2024
- Python, Scikit-learn, Pandas
- Analysed key drivers of employee turnover through EDA and clustering; applied SMOTE for class imbalance, k-fold cross-validation for model robustness, and identified the best-performing classifier with targeted retention recommendations.

EDUCATION

MSc Artificial Intelligence | Munster Technological University, Cork Sep 2025 – Aug 2026 (expected)

B.Tech | QIS Institute of Technology, Andhra Pradesh, India Jun 2018 – Jun 2022

GPA: 7.44/10

CERTIFICATIONS

Full Stack Development Certification | NxtWave Jun 2022 – Aug 2023

AI-101 Certification | Talent Next Jan 2024

SQL and Database Management | NxtWave Feb 2024